

1. Consider the code, $\mathcal{C}$, over $\mathbb{F}_2$, with parity-check matrix given by

$$H = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}.$$

- Construct the standard array for this code.
 - Use the standard array to decode the received vectors, $\mathbf{y} = (1, 0, 0, 1)$ and $\mathbf{y} = (1, 1, 1, 1)$.
 - What is the probability of decoding error if standard array decoding is used over a BSC with cross-over probability p ?
2. Prove that the minimum distance of a perfect code must be odd.
3. Show that the only possibly binary MDS codes of length n are $\{0, 1\}^n$, the repetition code, and the simple parity-check code.

[Hint: Assume that the parity-check matrix H is of the form $[I_{n-k} | A]$, for some A . Use the fact that the minimum distance, d , is the largest integer such that every set of $d - 1$ columns of H is linearly independent, to obtain the structure of A .]

4. Let $\mathcal{C}$ be an (n, M, d) code over a finite field $\mathbb{F}_q$. The Hamming distance of a vector $\underline{y} \in \mathbb{F}_q^n$ from $\mathcal{C}$, denoted by $d(\underline{y}, \mathcal{C})$, is defined as the Hamming distance between $\underline{y}$ and a nearest codeword in $\mathcal{C}$ to $\underline{y}$, that is,

$$d(\underline{y}, \mathcal{C}) = \min_{\underline{c} \in \mathcal{C}} d_H(\underline{y}, \underline{c}).$$

The covering radius of $\mathcal{C}$, denoted by r , is the largest distance from $\mathcal{C}$ of any vector in $\mathbb{F}_q^n$, that is ,

$$r = \max_{\underline{y} \in \mathbb{F}_q^n} d(\underline{y}, \mathcal{C})$$

- Show that if $\mathcal{C}$ is a linear (n, k, d) code over $\mathbb{F}_q$, then r is the largest among the Hamming weights of the coset leaders of $\mathcal{C}$ in $\mathbb{F}_q^n$.
 - Show that if $\mathcal{C}$ is a linear (n, k, d) code over $\mathbb{F}_q$ and H is an $(n - k) \times n$ parity check matrix of $\mathcal{C}$, then r is the smallest nonnegative integer such that every vector in $\mathbb{F}_q^{n-k}$ can be expressed as a linear combination of upto r columns in H .
5. Compute an upper bound on the number of codewords in a double-error-correcting binary block code of length 8.